

Drug Summary

Atezolizumab (Tecentriq) is a humanized IgG1 monoclonal antibody that targets the PD-L1 immune checkpoint, restoring anti-tumor T-cell activity ¹ ². In breast cancer, its use has been focused on triple-negative breast cancer (TNBC), which often exhibits high PD-L1 expression and tumor-infiltrating lymphocytes (TILs) ³. Atezolizumab combined with nab-paclitaxel demonstrated improved progression-free survival in PD-L1-positive metastatic TNBC (IMpassion130 trial) ⁴ ⁵, leading to an accelerated FDA approval in this subgroup. This benefit was pronounced in PD-L1-expressing tumors (assessed on immune cells) ⁶, reflecting the drug's mechanism of unleashing pre-existing anti-tumor immunity. However, a confirmatory trial (IMpassion131) did not show added benefit in the same setting, and the TNBC indication was subsequently withdrawn in the US in 2021 ⁷ ⁸. Outside of TNBC, Atezolizumab is not routinely used for other breast cancer subtypes due to low immunogenicity and lack of proven efficacy, though it continues to be explored in clinical trials. In current practice and guidelines, pembrolizumab has largely supplanted atezolizumab for PD-L1-positive TNBC, but atezolizumab remains an important case study in immunotherapy for breast cancer ⁹. Overall, atezolizumab's role in breast cancer is to enable the patient's immune system to recognize and destroy tumor cells, particularly in aggressive, immune-responsive TNBC subtypes ¹⁰ ³.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL3707227 ¹¹
- **DrugBank ID:** DB11595 ¹²
- **Synonyms/Trade Names:** Atezolizumab is marketed as **Tecentriq** (IV formulation) and **Tecentriq Hybryza** (subcutaneous formulation with hyaluronidase) ¹³. It has been referred to colloquially as “Atezo.” Development codes include **MPDL3280A** (also written as MPDL 3280A) and **RG7446**, which were used during clinical development ¹⁴. These synonyms are all explicitly associated with atezolizumab ¹⁵ ¹⁶. For example, MPDL3280A was the Genentech code name, and Tecentriq® is the approved brand name in the US and EU ¹⁵ ¹⁶.

Mechanism of Action (Breast Cancer Context)

Atezolizumab's mechanism of action is blockade of the PD-L1 (CD274) protein on tumor cells and immune cells, preventing PD-L1 from engaging PD-1 on T-cells ¹. In breast cancer – especially TNBC – many tumors exploit the PD-1/PD-L1 axis to suppress cytotoxic T-lymphocyte activity. By binding PD-L1, atezolizumab “releases the brakes” on the immune system, allowing T-cells to recognize and kill cancer cells ¹. This is particularly relevant in TNBC, which often has a higher mutational burden and TIL presence, making it more immunogenic than hormone receptor-positive or HER2+ breast cancers ³. In PD-L1-positive TNBC, atezolizumab plus chemotherapy can induce an “**immune-inflamed**” **tumor microenvironment**, characterized by elevated interferon-gamma signaling and T-cell activation ¹⁷ ¹⁸. Mechanistically, the drug does not directly kill tumor cells; rather, it **revives anti-tumor immunity**. This can translate into tumor shrinkage and delayed progression, as seen in clinical trials ⁴ ¹⁰. Subtype nuances: HR+/HER2- breast cancers typically have lower baseline TILs and PD-L1 levels, and have not shown

significant benefit from atezolizumab in trials. HER2-positive tumors can express PD-L1, but combination immunotherapy strategies are still investigational. The **most robust activity has been in TNBC**, especially the basal-like, immune-activated subset (BLIA), where pre-existing immune response is high ¹⁹. In contrast, basal-like immune-suppressed (BLIS) TNBC – often characterized by high TGF- β and fibroblast activity – responds poorly, as an immunosuppressive stroma can exclude T-cells despite PD-L1 blockade ¹⁹. Overall, atezolizumab's efficacy in breast cancer hinges on tumor immune contexture: it is effective when an anti-tumor immune response is present but restrained by PD-L1, and ineffective when tumors lack T-cell infiltration or have alternate immune evasion pathways.

Primary Targets (Human)

Atezolizumab's direct molecular target is **PD-L1** (programmed death-ligand 1), which is encoded by the gene **CD274** ². PD-L1 is an immune checkpoint ligand expressed on tumor cells and tumor-infiltrating immune cells; it binds the PD-1 receptor on T cells to inhibit their activity. By targeting PD-L1, atezolizumab prevents it from engaging PD-1 and also from interacting with CD80 (B7-1) on T cells, thereby lifting inhibitory signals on T cells ¹. **CD274 (PD-L1)** is thus the primary and only intended target. There are no enzymatic or cytosolic targets – the drug functions via this cell-surface immune modulatory target. Atezolizumab's specificity for human PD-L1 is high, with no known off-targets in humans; binding has been confirmed in crystal structure analysis of the PD-L1/atezolizumab complex ²⁰.

Pathways (Overview)

Atezolizumab's impact in breast cancer is closely tied to immune-related pathways and compensatory tumor signaling pathways. Key pathways affected include:

- **PD-1/PD-L1 Checkpoint Pathway (KEGG hsa05235) – Immune checkpoint signaling:** The direct pathway of action; blocking PD-L1 leads to activation of T-cell mediated immunity ²¹. Upregulation of this pathway (i.e. tumor PD-L1 expression) is required for the drug to have a target; its blockade converts an immune-suppressed environment into an immune-active one.
- **Interferon-Gamma Response (MSigDB Hallmark: HALLMARK_INTERFERON_GAMMA_RESPONSE) – Immune activation signature:** Responding tumors often show enriched IFN- γ signaling, which includes antigen presentation and chemokine release that recruits T cells ²². This pathway is typically **upregulated in sensitivity** contexts, as IFN- γ from T-cells induces PD-L1 and primes tumor cells for immune attack.
- **T-cell Activation/Allograft Rejection Signature (MSigDB: HALLMARK_ALLOGRAFT_REJECTION) – Cytotoxic lymphocyte infiltration:** Effective treatment is associated with gene sets for T-cell and NK cell activity ²². High baseline or on-treatment upregulation of these immune pathways correlates with sensitivity (a so-called “hot” tumor).
- **NF- κ B/TNF α Signaling (Hallmark_TNFA_SIGNALING_VIA_NFKB) – Inflammatory cytokine pathway:** Checkpoint blockade can be more effective in tumors with an ongoing inflammatory response. TNF α /NF- κ B-related gene signatures can indicate an immune-inflamed tumor microenvironment ¹⁷. Upregulation might reflect productive inflammation aiding T-cell function.
- **IL-6/JAK-STAT3 and IL-2/STAT5 Signaling (Hallmarks) – Immune cytokine signaling:** These pathways can shape the tumor immune microenvironment. For instance, IL2-STAT5 signaling is essential for

effector T cell proliferation, and its presence may enhance responses ¹⁷ . Conversely, aberrant IL-6/STAT3 activation often promotes immune suppression; its downregulation may be beneficial.

- **Epithelial-Mesenchymal Transition (EMT) and TGF- β Signaling (Hallmark_EMT, TGF β pathway)** – *Mesenchymal/stromal pathway*: Atezolizumab exposure in TNBC has been shown to **downregulate EMT-related genes** and TGF- β -driven processes ²³ ²⁴ . EMT and high TGF- β create a fibrotic, T-cell-excluding stroma (the “immune-excluded” phenotype). Thus, **downregulation of EMT/TGF- β pathways** is associated with greater sensitivity, whereas their upregulation confers resistance.
- **Hypoxia and Angiogenesis (Hallmark_HYPOXIA, Hallmark_ANGIOGENESIS)** – *Tumor microenvironment stress*: PD-L1 blockade in vitro was found to *reduce the induction of hypoxia* and its gene signatures ²⁵ ²⁶ . Hypoxia can suppress immune function; thus, lowering hypoxia (via better perfusion or less HIF1 α activity) may improve immune attack. High hypoxia/angiogenesis pathways, on the other hand, correlate with immune suppression and resistance.
- **Cell Cycle Proliferation (E2F targets, G2M checkpoint – Hallmarks)** – *Proliferation and DNA replication*: Atezolizumab's action can indirectly slow tumor cell proliferation by immune-mediated killing. In responders, gene sets like E2F targets and G2/M checkpoint tend to be *downregulated*, reflecting decreased proliferation ²⁷ . High proliferation signatures at baseline (often seen in aggressive tumors) don't preclude response, but their reduction on-treatment is a marker of effective therapy.
- **DNA Repair and Genomic Instability – DDR pathways**: Interestingly, in TNBC cell models, PD-L1 blockade *upregulated DNA repair genes and promoted DNA repair fidelity* ²⁵ . This could be a cell-intrinsic response to immune attack-induced DNA damage. From a clinical angle, tumors with **DNA repair deficiency (HRD)** often have high mutation load and TILs, making them more sensitive to immunotherapy ²⁸ . Thus, DNA repair pathways play a nuanced role: **downregulated DNA repair (HRD)** can increase initial sensitivity (due to immunogenicity), whereas on-treatment, some DNA repair upregulation might be a tumor cell survival response.
- **PI3K/AKT/mTOR Pathway – Oncogenic survival pathway**: This pathway can modulate immune responses; PTEN loss and PI3K activation in tumors have been linked to an immune-cold microenvironment and resistance to PD-1/PD-L1 blockade ²⁹ . There is ongoing research combining AKT inhibitors with atezolizumab to overcome this resistance mechanism ²⁹ . Downregulation or inhibition of PI3K/AKT signaling can thus **sensitize tumors**, while upregulation (e.g. via PTEN loss) is associated with immune evasion.
- **Wnt/ β -Catenin Pathway – Immune evasion via tumor/stroma signaling**: Although not unique to breast cancer, Wnt/ β -catenin activation in tumors or CAFs can exclude T-cells and induce PD-L1 expression in the microenvironment ³⁰ . In TNBC, Wnt pathway upregulation (e.g. in cancer stem-like cells or fibroblasts) correlates with poor response to checkpoint inhibitors. **Inhibiting Wnt** can drive a more T-cell-permissive environment and restore sensitivity to atezolizumab ³⁰ .

These pathways illustrate how atezolizumab's efficacy is a balance of **immune activation** pathways (IFN γ , T-cell activation, etc.) versus **tumor-intrinsic or microenvironmental escape** pathways (EMT/TGF- β , Wnt, PI3K, etc.). Each pathway may be **upregulated or downregulated** in response to therapy or as a baseline tumor characteristic, influencing whether the tumor is sensitive or resistant to PD-L1 blockade.

Upregulated Pathways (Context/Subtype)

Certain pathways are transcriptionally upregulated in response to atezolizumab or are found at higher levels in responding tumors. These can contribute to drug sensitivity or, in some cases, an adaptive resistance. Key upregulated pathways and contexts:

- **Interferon Signaling Pathways (Type II IFN – IFN γ):** In responders (especially PD-L1-positive TNBC), IFN- γ related pathways are upregulated upon treatment. For example, in the atezolizumab+nab-paclitaxel arm of IMpassion130, on-treatment biopsies showed increased expression of IFN γ -responsive genes and PD-L1 itself ³¹ ³². This IFN γ pathway upregulation is predominantly a marker of *sensitivity*, as it indicates active engagement of immune cells with the tumor. Notably, IFN γ release induces more PD-L1 on tumor and immune cells (a feedback which atezolizumab can counteract) and upregulates antigen presentation machinery. In vitro, treating TNBC cells with atezolizumab led to elevated apoptotic signals and immune-related transcripts, consistent with IFN-mediated effects ²³ ²⁴. **Subtype context:** This upregulation is mainly seen in immunogenic TNBC; HR+/HER2- tumors rarely mount such an IFN γ response due to fewer baseline TILs.
- **T-cell Activation and Cytotoxic Pathways:** Genes and pathways related to CD8+ T cell activation, granzyme/perforin (cytotoxic function), and Th1-polarization are upregulated in responders. A hallmark example is the **Allograft Rejection gene set**, which was significantly enriched (upregulated) in tumors responding to atezolizumab ²² ³³. This reflects increased presence and activity of cytotoxic T-lymphocytes. Similarly, **Inflammatory Response** pathways (e.g. involving IL-1, TNF α) become elevated, indicating an “inflamed” microenvironment conducive to tumor killing ¹⁷. **Subtype context:** Upregulation of these pathways is most notable in TNBC (particularly basal-like immune-activated subtype) ¹⁹. In luminal (HR+) cancers, these signatures are generally low and remain low (indicative of poor immune activation).
- **PD-L1 Expression (on Immune Cells and Tumor Cells):** Paradoxically, effective treatment can transiently increase PD-L1 levels in the tumor microenvironment – an expected pharmacodynamic effect of T-cell reactivation. In patients who achieved responses (CR/PR) in IMpassion130, an increase in PD-L1+ tumor-infiltrating immune cells was observed after 4 weeks of therapy ³¹. This is driven by IFN γ from activated T cells and signifies engagement of the immune system. While PD-L1 upregulation is a tumor’s adaptive resistance mechanism, the presence of this dynamic suggests the drug is hitting its target and is usually associated with tumor regression when accompanied by immune cell infiltration ³⁴ ³⁵. (Clinically, this is why PD-L1 is both a biomarker for patient selection and part of the immune feedback loop.)
- **Apoptosis Pathways:** Atezolizumab does not directly induce tumor apoptosis, but upregulation of pro-apoptotic pathways in tumor cells can occur secondary to immune attack. In TNBC cell studies, PD-L1 blockade led to increased expression of pro-apoptotic genes and higher rates of apoptosis/necrosis in tumor cells ³⁶. Pathway analysis showed **upregulation of genes promoting apoptosis** (e.g. BIM, FAS) and **downregulation of anti-apoptotic signals** in treated cells ³⁷ ³⁸. In patients, an on-treatment increase in apoptotic gene signatures within tumors would align with effective immune-mediated killing. This upregulated “apoptosis execution” pathway is desirable and correlates with sensitivity.

- **DNA Damage Response (DDR) Activation:** Counterintuitively, PD-L1 blockade in vitro was reported to *upregulate certain DNA repair pathways* in TNBC cells ²⁵ . Specifically, KEGG analysis showed enrichment of **p53 pathway genes** among upregulated transcripts post-atezolizumab ³⁹ . This suggests that, as tumor cells come under immune attack (producing DNA damage via ROS and perforin/granzyme), they attempt to activate p53-mediated DNA repair and cell-cycle arrest. While this is a tumor cell survival response, its presence (upregulated p53/DDR signaling) could indicate that the immune assault is occurring (hence a marker of *ongoing response*). However, if too effective, it might allow tumor cells to recover – thus DDR upregulation is a double-edged sword. In a clinical context, baseline *deficiency* in DNA repair (HRD mutations) is associated with immunogenicity ²⁸ , but once treatment starts, tumor cells that can mount a DNA damage response may survive longer. Net effect: **upregulation of p53/DDR upon therapy reflects active engagement (sensitivity) but could contribute to eventual resistance if those cells repair and survive.**

- **Sirtuin and Metabolic Pathways:** Gene set analyses (e.g. Ingenuity IPA) from lab studies pointed to changes in metabolic regulators like the **Sirtuin signaling pathway** upon PD-L1 blockade ²⁴ . Sirtuin pathways (involving SIRT1, etc.) intersect with metabolism, stress responses, and DNA repair. In MDA-MB-231 cells, atezolizumab altered these pathways, possibly promoting a more oxidative and less glycolytic metabolism (since interferon and immune attack can cause metabolic stress on tumor cells). Upregulation of oxidative phosphorylation genes was noted in responders' tumors in some transcriptional analyses ²⁷ ⁴⁰ . An increase in oxidative metabolism and ROS handling might be a tumor adaptation to immune attack. While not a classical “sensitivity” pathway, these metabolic shifts indicate an ongoing tumor-immune interaction.

Summary: Upregulated pathways in the context of atezolizumab therapy largely align with a vigorous immune response: interferon signaling, T-cell activation, inflammatory cytokines, and apoptotic processes are all heightened in sensitive tumors ²² ³¹ . These changes are most prominent in immunogenic subtypes of breast cancer (TNBC). They provide a mechanistic rationale for combination strategies as well – for example, IFN γ pathway upregulation inducing PD-L1 could be countered by continued checkpoint blockade, and elevated T-cell activation might be further enhanced by co-stimulatory agonists or IL-2 therapy. Importantly, the presence of these upregulated immune pathways is a **favorable sign** for patients on atezolizumab.

Downregulated Pathways (Context/Subtype)

Atezolizumab exposure and effective anti-tumor immune responses are also associated with the **downregulation** of certain pathways in breast tumors. These pathways are often those that promote tumor growth, metastasis, or immune evasion. Notable downregulated pathways include:

- **EMT and Metastasis-Related Pathways:** One of the clearest transcriptional effects observed in TNBC models is the suppression of epithelial-mesenchymal transition (EMT) programs upon PD-L1 blockade ²³ ³⁷ . Genes driving EMT and invasion (e.g. *VIM* (vimentin), *ZEB1*, *TWIST*, or noted examples IL11RA, FOXQ1, KLF8) were **significantly downregulated** after atezolizumab treatment in a TNBC cell line ⁴¹ ²³ . Concurrently, genes that maintain the epithelial state or inhibit metastasis (ESRP2, etc.) were upregulated ²³ . This suggests PD-L1 signaling in tumor cells might contribute to EMT and aggressive phenotype, and blocking it reverses some of that. In patients, responders show lower expression of EMT and TGF- β associated gene signatures on-treatment compared to baseline

or non-responders ¹⁹. **Subtype context:** Basal-like immune-suppressed (BLIS) TNBC is characterized by high EMT and TGF- β signaling; these tumors are relatively resistant ¹⁹. Atezolizumab's benefit is limited in such cases, correlating with the idea that if EMT/TGF- β pathways remain high (not downregulated), response is poor. Thus, **downregulation of EMT/metastatic pathways** (either due to direct effects or indirectly via immune pressure) is linked to sensitivity.

- **Hypoxia and Angiogenesis:** Atezolizumab-treated TNBC cells showed *reduced induction of hypoxia-responsive genes* ²⁵. Moreover, analysis of tumor biopsies has indicated that non-responding tumors often have high hypoxia/angiogenic signatures (VEGF-related pathways) that persist, whereas responding tumors shift to a less hypoxic, more inflamed state ¹⁷ ⁴². Hypoxia can impede immune effector function and promote an immunosuppressive metabolism. **Downregulation of hypoxia pathways** (and possibly normalization of vasculature) in the tumor microenvironment can enhance immune cell infiltration and is associated with better outcomes. For example, patients whose tumors had lower expression of HIF1 α target genes on-treatment had improved PFS in one correlative study (suggesting that effective therapy alleviated or overcame hypoxia) ¹⁷.
- **Cell Cycle/Proliferation Pathways:** As the immune system attacks the tumor, one expects to see a reduction in tumor proliferation. Indeed, hallmark gene sets for **E2F targets, G₂-M checkpoint, and MYC-driven proliferation** were found to be *downregulated* in responders vs non-responders in IMpassion130 correlative analyses ²⁷ ⁴³. Essentially, tumors that respond slow their cell cycle (due to immune-mediated cell death and interferon-induced cell-cycle arrest via pathways like p21). In contrast, tumors that are resistant often maintain high proliferation signatures. Thus, a drop in Ki-67 and related gene expression (downregulation of proliferation pathways) is a marker of effective therapy. **Subtype context:** TNBC generally has high proliferation at baseline; a steep on-treatment decrease in proliferation is a good sign. HR+ tumors with low proliferation to begin with have other dominant resistance factors (lack of immune engagement), so proliferation change is less informative there.
- **Anti-Apoptotic and Survival Signaling:** Corresponding to increased apoptosis signals, many **pro-survival pathways are suppressed** by successful therapy. For instance, the NF- κ B pathway can have tumor-promoting, anti-apoptotic effects; while parts of NF- κ B are pro-immune, tumor-intrinsic NF- κ B survival signaling (driving IL-8, BCL-XL, etc.) was noted to decrease with atezolizumab in lab studies ³⁷ ³⁸. Similarly, **BCL2/BCL-xL** gene expression might fall, and pathways like **PI3K/AKT** (which provide survival signals) can be curtailed if immune attack is effective (either via IFN-induced senescence or by direct killing of those cells). Indeed, in responders, metabolic and growth factor pathways (like mTOR signaling, fatty acid synthesis, etc.) trended downward, consistent with a starvation of the tumor as resources are diverted to immune and stress responses ⁴⁴ ²⁷.
- **Immunosuppressive Cytokine Networks:** Downregulation of pathways involving immunosuppressive cytokines or cells is also observed. For example, if therapy is effective, one might see reduced expression of **IL-10, TGF- β** , or markers of regulatory T-cells and M2 macrophages. In the Roche analysis, **stromal signatures (many TGF- β driven) were substantially decreased in patients with progression on both arms** ⁴⁵, suggesting that those who did poorly had persistent stroma; by contrast, those who responded could have a relative decrease in these immunosuppressive stromal signals (though data showed that in progression, immunity drops rather than stroma drops). Specifically, *TGF- β signaling* in immune-excluded tumors remains high unless countered. Combination approaches to intentionally downregulate TGF- β (e.g. TGF- β

inhibitors) are being explored to synergize with atezolizumab ⁴⁶. So while single-agent atezo may not fully suppress TGF- β , **the relative downregulation of TGF- β -related gene sets** is associated with an “inflamed” phenotype and better response.

Summary: Downregulated pathways with atezolizumab therapy primarily involve those that the tumor uses to thrive and hide: EMT/metastasis programs, proliferative drive, hypoxic tolerance, and immunosuppressive mediators are diminished in successful treatment scenarios ⁴⁷ ³⁷. These changes tilt the balance toward effective immune-mediated tumor control. Notably, many of these downregulated processes (EMT, angiogenesis, TGF- β signaling) are interrelated and define the immune-excluded tumor microenvironment. Their suppression is both a consequence of and a facilitator for immune infiltration. In practice, observing these pathway changes has spurred combination strategies – e.g., adding TGF- β blockade to increase T-cell penetration, or using angiogenesis inhibitors to normalize vessels and reduce hypoxia, thereby enhancing checkpoint inhibitor efficacy.

Sensitivity Pathways (Mechanisms of Sensitivity)

Mechanisms of sensitivity to atezolizumab in breast cancer revolve around pathways and biomarkers that **permit or enhance immune system attack** on the tumor. These include genetic alterations, gene expression patterns, or microenvironment features that make the tumor more vulnerable to PD-L1 blockade. We detail them by pathway context, emphasizing whether upregulation or downregulation of each pathway increases sensitivity:

Sensitivity Pathways for Upregulation: These are pathways whose higher activity or expression correlates with **increased sensitivity** to atezolizumab, meaning their presence helps the drug work better:

- **Interferon- γ -Related Pathway Upregulation:** Perhaps the most critical sensitivity marker is an ongoing *interferon- γ (IFN γ) immune response* in the tumor. Upregulation of an IFN γ gene signature (e.g. CXCL9, CXCL10, IDO1, MHC molecules) indicates a T-cell-inflamed tumor that is likely to respond ²². In TNBC, a high IFN γ signature (often part of the “T-cell inflamed” or “hot” tumor phenotype) is associated with better outcomes on PD-1/PD-L1 inhibitors ²². IFN γ upregulates PD-L1 on tumor and immune cells, thus creating the dependency that atezolizumab exploits; it also upregulates antigen presentation (HLA class I/II) on tumor cells ⁴⁸. In short, tumors that have *already upregulated* IFN γ pathways (e.g. from endogenous anti-tumor immunity) are highly sensitive to PD-L1 blockade. This is evidenced by analyses of IMpassion130 where an “IFN γ 18-gene signature” was prognostic for response ²² (this signature overlaps with the MSigDB HALLMARK_INTERFERON_GAMMA_RESPONSE gene set). **Subtype:** Upregulated IFN γ pathways are mainly seen in TNBC (especially BLIA subtype) and some immune-rich HER2+ cancers; they are rarely intrinsic to HR+ cancers.
- **T-cell Activation and Cytotoxic Function (Up):** High baseline levels or induction of cytotoxic T lymphocyte (CTL) pathways correspond to sensitivity. This includes upregulation of pathways such as **Granzyme B/Perforin-mediated killing, Th1 differentiation, and CD8 effector function**. If these pathways are active, it means T-cells are present and poised to kill – they just need the checkpoint released. Clinically, *high CD8+ T-cell gene signatures* or TIL counts in TNBC predict better response to atezolizumab ³. In trials, patients with “immune-inflamed” tumors (lots of CD8, granzyme, IFN γ , etc.) had significantly longer PFS on atezo ³². Thus, upregulation of CTL activation pathways (such as the **Allograft Rejection signature or T-cell receptor signaling gene sets**) is a sensitivity

mechanism. **Mechanistic rationale:** These indicate an ongoing anti-tumor immune response that was being inhibited by PD-L1; once released by atezolizumab, the T-cells can perform tumor cell lysis effectively.

- **Chemokine Signaling for Lymphocyte Trafficking (Up):** Tumors that upregulate certain chemokines are more sensitive because they attract immune cells. For example, **CXCL9, CXCL10, CCL5 upregulation** (often driven by IFN γ and NF- κ B) recruits T cells and dendritic cells into the tumor. High expression of these chemokines in TNBC is associated with robust immune infiltration and has been linked to better responses to checkpoint blockade ²². In essence, an upregulated “immune recruitment” pathway makes the tumor a better target for immunotherapy. Conversely, tumors lacking chemokine signals remain cold. So, pathways like **NF- κ B (for cytokine production) and TBK1/IRF3 (for Type I IFNs)**, if upregulated, enhance sensitivity by orchestrating immune cell entry.
- **BRCA/HRD-Related Neoantigen Creation:** While not a pathway per se, the *consequence* of homologous recombination deficiency (HRD) – namely genomic instability leading to neoantigens – is a sensitivity factor. TNBC with germline BRCA1/2 mutations or high HRD scores often have more mutations and TILs ²⁸. Upregulation of DNA damage (paradoxically) can increase tumor immunogenicity. In other words, tumors that accumulate DNA damage (due to HRD) effectively “upregulate” the mutation-generated antigenic pathways (like the MSigDB **HALLMARK_DNA_REPAIR** being low means more damage). These tumors tend to respond better to atezolizumab, especially when combined with DNA-damaging chemo. For instance, subsets of patients with BRCA1/2-mutated TNBC showed numerically higher response rates to checkpoint inhibitors in studies ²⁸. Thus, from a pathway perspective: *upregulation of the consequences of DNA damage (like STING pathway activation by cytosolic DNA)* can improve sensitivity. Indeed, cGAS-STING (innate sensing of DNA) activation leads to type I IFN production and T-cell recruitment, so tumors that naturally activate STING (due to DNA damage) are more likely to be sensitive.
- **Immune Checkpoint Ligands on Tumor (PD-L1 itself and others):** Counterintuitive but important – tumors that highly express PD-L1 (i.e. *upregulate PD-L1*) are required for atezolizumab to work (no target otherwise). PD-L1 expression on tumor or immune cells is the predicate for response ⁶. That’s why PD-L1 IHC-positive patients benefitted in IMpassion130. Upregulation of PD-L1 often co-occurs with T-cell inflammation (via IFN γ). Similarly, some data suggest tumors with high **PD-L2** (another PD-1 ligand) might also benefit from PD-1/L1 axis blockade (though atezo doesn’t bind PD-L2, PD-L2 is not a dominant factor in breast cancer). The key is that PD-L1-high status ($\geq 1\%$ IC by SP142 assay, for example) was needed for regulatory approval ⁶. So, while PD-L1 upregulation is a tumor immune-evasion tactic, it is paradoxically a *marker of sensitivity* since it identifies tumors where the immune system is engaged and just needs checkpoint release.
- **Certain HLA and Antigen Presentation Genes (Up):** High expression of HLA class I and II molecules on tumor cells and antigen-presenting cells facilitates T-cell recognition. If a pathway like **“Antigen Processing and Presentation” (GO:0019882)** is upregulated, that means tumor peptides are being presented to T-cells effectively – a necessary component for response. In breast cancer, especially TNBC, higher MHC-I gene expression correlates with TIL presence and better immunotherapy response (conversely, loss of MHC is a resistance mechanism). So, an *upregulated antigen presentation pathway* (due to IFN γ or intrinsic factors) is a pro-sensitivity factor. For example, one TNBC case study found that an acquired JAK2 mutation (which downregulated MHC expression via

IFN signaling loss) led to resistance ⁴⁹ ; by contrast, tumors with intact/upregulated JAK-STAT and MHC pathways responded well.

Sensitivity Pathways for Downregulation: These are pathways where a *decrease* in their activity makes the tumor more sensitive to atezolizumab. In other words, if these pathways are suppressed (either inherently or via combination therapy), the efficacy of PD-L1 blockade is increased:

- **Wnt/ β -Catenin Pathway (Down):** Downregulating the Wnt/ β -catenin signaling in tumors or the tumor microenvironment can greatly improve sensitivity to immunotherapy. Wnt pathway activation in tumors leads to exclusion of T-cells (through effects on dendritic cells and T-cell priming). In TNBC, studies have shown that Wnt signaling contributes to an immune-cold phenotype; for instance, TNBC cells with “stem-like” properties have active Wnt signaling and high PD-L1, mediating immune evasion ⁵⁰ . In a mouse model, *inhibition of Wnt* (pharmacologically or genetically) led to increased T-cell infiltration and **sensitized previously resistant tumors to PD-L1 blockade** ^{30 51} . Therefore, **downregulation of the Wnt/ β -catenin pathway** removes a block on immune surveillance, converting the TME into one more permissive for atezolizumab action. Clinically, this suggests that patients with low Wnt pathway activity (or those treated with Wnt inhibitors) could respond better. Wnt is not routinely measured in patients, but its downstream effects (e.g. absence of T-cells in β -catenin-active tumors) are well documented in melanoma and likely apply to TNBC as well.
- **TGF- β Signaling (Down):** High TGF- β levels in TNBC create a dense stroma and exclude immune cells ^{46 49} . If TGF- β pathway activity is *downregulated*, the physical and immunosuppressive barrier is lifted. Indeed, in an exploratory analysis of IMpassion130, patients with an “**immune-infiltrated**” phenotype (low TGF- β , high TILs) derived more benefit than those with an “**immune-excluded**” phenotype (high TGF- β , collagen, fibroblasts) ¹⁹ . External evidence from other tumors (e.g. lung) shows that combining TGF- β blockade with anti-PD-L1 can convert exclusion to infiltration, improving efficacy. Thus, tumors where TGF- β signaling is intrinsically low (e.g. lack of TGF- β production, or mutations that attenuate TGF- β receptors) are more sensitive to atezolizumab. Downregulation of TGF- β effects – such as reduced fibroblast activation and extracellular matrix deposition – correlates with better immune cell penetration and response. **Mechanistically:** TGF- β suppresses cytotoxic T-cell function and promotes Tregs and myeloid suppressors; removing that influence with pathway downregulation allows a more robust anti-tumor immunity to proceed alongside PD-L1 blockade ⁴⁶ .
- **PI3K/AKT Pathway (Down):** Tumors with loss of PTEN (hence up PI3K signaling) have been linked to poorer response to immunotherapy (as noted, PTEN loss can lead to fewer TILs due to tumor-intrinsic immune resistance) ²⁹ . Conversely, downregulating the PI3K/AKT pathway – for example, using an AKT inhibitor (ipatasertib) – was shown in a trial (Ipatasertib+Atezolizumab) to potentially augment immune response in TNBC ²⁹ . Preclinically, inhibiting PI3K γ in myeloid cells or PI3K α/δ in tumors can reduce immunosuppressive cytokines and increase T-cell activity. Therefore, **low PI3K pathway activity** (whether inherently or via a drug) makes the tumor microenvironment less immunosuppressive, boosting sensitivity to PD-L1 blockade. One reason is that active AKT in tumors can lead to production of VEGF and CXCLs that recruit suppressive myeloid cells; if that is dampened, the immune cells function better. So, **downregulation of PI3K/AKT** is a pro-sensitivity condition.
- **IDO/Kynurenine Pathway (Down):** Indoleamine-2,3-dioxygenase (IDO1) is an immune regulatory enzyme that tumors and immune cells use to deplete tryptophan and suppress T-cells. While not

specific to breast cancer, TNBC can express IDO1 especially after IFN γ exposure (which can be a negative feedback) ⁵². High IDO activity correlates with resistance to immunotherapy in some settings. Thus, **downregulating IDO1** (or pharmacologically inhibiting it) can enhance T-cell responses to PD-L1 blockade. In fact, trials combining IDO inhibitors with checkpoint blockers were based on this rationale. In breast cancer, patients with lower IDO1 expression at baseline (or greater drop on treatment) might have better T-cell persistence. This is more speculative because clinical validation in breast is limited, but conceptually, **loss of IDO-mediated suppression** is sensitizing.

- **Regulatory T-Cell and MDSC Pathways (Down):** Atezolizumab works better if the pathways sustaining Tregs and myeloid-derived suppressor cells (MDSCs) are kept in check. For example, tumors with lower expression of *CCR4* (Treg attractant), *ARG1* or *IL-6* (MDSC support) might respond better. Downregulating those immunosuppressive cell-recruiting pathways essentially **reduces the brakes** on the immune system beyond PD-L1. While PD-L1 blockade itself doesn't directly target Tregs or MDSCs, a favorable scenario is when, either by tumor-intrinsic lack or by combination therapy, those suppressive influences are minimal. In TNBC, a high ratio of T-effector to Treg is a biomarker of response – which can be achieved by having fewer Tregs (i.e. downregulated Treg pathways). Agents like CSF1R inhibitors (targeting tumor-associated macrophages) or CXCR2 inhibitors (blocking neutrophil/MDSC chemotaxis) effectively *downregulate suppressor cell pathways* and have been hypothesized to sensitize tumors to PD-L1 blockade.

In summary, **sensitivity-enhancing pathways** often involve an *active immune attack (which can be further unleashed)* or the *absence of major immune evasion circuits*. Upregulating pro-immune pathways (IFN γ , T-cell activation) and downregulating tumor/stromal escape pathways (Wnt, TGF- β , PI3K) both tilt the balance in favor of atezolizumab efficacy. These insights underscore why patient selection (e.g. via PD-L1 and TIL status) and combination therapies (e.g. adding TGF- β or AKT inhibitors) are crucial in the clinical use of atezolizumab.

Resistance Pathways (Mechanisms of Resistance)

Despite initial responsiveness in some breast cancers, various pathways can mediate **resistance** to atezolizumab. Resistance may be primary (innate, where the drug never works) or acquired (developing after initial response). Mechanisms of resistance include upregulation of alternate immune evasion pathways, genetic changes, or adaptive shifts in the tumor microenvironment. We outline proven or strongly suggested resistance mechanisms by pathway, again distinguishing those that confer resistance when *upregulated vs downregulated*:

Resistance Pathways for Upregulation: Pathways that, when increased in activity or expression, lead to immune evasion and treatment failure:

- **Alternative Immune Checkpoints (LAG-3, TIM-3, etc) Upregulation:** Tumors can escape PD-L1 blockade by upregulating other inhibitory receptors on T-cells. For instance, studies in TNBC show that **blocking PD-1/PD-L1 leads to compensatory upregulation of LAG-3 and TIM-3 on T cells** ⁵³ ⁵⁴. LAG-3 (Lymphocyte Activation Gene-3) and TIM-3 are checkpoint molecules that, when overexpressed, can exhaust T cells even in the absence of PD-1 signaling. Clinically, in patients who progress on atezolizumab, TILs often express higher LAG-3. This has driven trials of combined PD-(L)1 and LAG-3 blockade. Thus, **upregulation of LAG-3/TIM-3 pathways** is a resistance mechanism:

it provides redundant brakes on T-cells. Similarly, tumor or immune cell upregulation of **CTLA-4** (less documented in breast than in melanoma) could also contribute to resistance, though CTLA-4 is more relevant in priming phase.

- **Immunosuppressive Enzymes/Cytokines (IDO, IL-10, VEGF) Upregulation:** Tumors can secrete factors that blunt immune attack. **IDO1** upregulation, as mentioned, creates a tryptophan-depleted, kynurenine-rich milieu that paralyzes T-cells. If a tumor under immune pressure increases IDO1 expression (often via IFN γ stimulus), it can locally suppress T-cell proliferation and function, allowing tumor survival despite PD-L1 being blocked ⁵². Another is **IL-10**, an anti-inflammatory cytokine; increased IL-10 from tumor or tumor-associated macrophages can dampen the immune response even if PD-L1 is blocked. **VEGF-A** is worth noting: its upregulation promotes angiogenesis and also directly inhibits T-cell function and trafficking. High VEGF correlates with poor response to immunotherapy, which is why combining anti-angiogenics (e.g. bevacizumab) with PD-L1 inhibitors is being studied (and in some cancers, approved). In summary, tumors that **upregulate soluble immunosuppressive pathways** (metabolic enzymes like IDO, cytokines like TGF- β /IL-10, or metabolites like adenosine via CD73 upregulation) exhibit resistance by creating a non-permissive microenvironment.
- **Wnt/ β -Catenin Pathway (Up):** As the converse of sensitivity, if Wnt signaling remains high or further increases, the tumor can resist T-cell infiltration. In TNBC, Wnt-active tumors often have a “cold” immune microenvironment. For example, an analysis identified that TNBC cells with high Wnt output were associated with low TIL counts and poor response to atezolizumab ⁵⁰. Upregulation of Wnt in **cancer-associated fibroblasts (CAFs)** is also problematic: Wnt-driven CAFs can adopt a myofibroblastic phenotype that secretes factors to exclude T cells. A key experimental finding: in a bone marrow-derived myofibroblast (BMF) co-culture model, **activation of Wnt/ β -catenin in CAFs led to PD-L1 upregulation on those CAFs and conferred resistance to anti-PD-L1 therapy**, whereas Wnt inhibition reversed this resistance ⁵⁵ ³⁰. Hence, persistent or increased Wnt/ β -catenin signaling in either tumor cells or stroma is a potent resistance mechanism (preventing the immune system from effectively entering or attacking the tumor).
- **PI3K/AKT/mTOR Pathway (Up):** If a tumor responds poorly, often the PI3K pathway is aberrantly active (through PTEN loss, PIK3CA mutation, etc). High AKT signaling can lead to production of immunosuppressive factors and can make tumor cells less reliant on PD-L1 for survival (since they have strong intrinsic survival signaling). In a clinical sense, presence of PTEN loss (which “upregulates” PI3K signaling) was linked to resistance to checkpoint blockade in other cancers, and TNBC often has PTEN loss. Moreover, **hyperactivation of AKT** can cause exclusion of T-cells by modulating tumor metabolism (e.g. acidifying the microenvironment via glycolysis). Indeed, a trial combining an AKT inhibitor aimed to counteract this: initial data indicated that patients whose tumors had PI3K pathway alterations were less responsive to immunotherapy alone ²⁹, underscoring that **upregulated PI3K/AKT** is a resistance pathway.
- **MYC Oncogene Pathway (Up):** MYC is a transcription factor that when amplified/overexpressed can drive numerous pro-tumor effects, including repression of immune responses. MYC can suppress antigen presentation and promote immunosuppressive metabolites. Preclinical studies in TNBC mouse models have shown that MYC-overexpressing tumors are highly immune-evasive, but combining checkpoint blockade with strategies that target MYC-driven processes (like bromodomain inhibitors or mTOR inhibitors) improved outcomes ⁵⁶. In essence, **upregulation of MYC** (and its

target genes) leads to increased PD-L1 expression (MYC can induce PD-L1 transcription) but also other immunoevasion tactics, resulting in only partial benefit from PD-L1 blockade. Unless MYC activity is curtailed, it can foster redundant escape mechanisms. Thus, high MYC activity is a resistance biomarker, and downregulating MYC-driven pathways (cell cycle, ribosome biogenesis) may be needed to fully restore immune control.

- **Stromal Fibrosis/Collagen (Up):** Not a classical “pathway,” but the extent of desmoplasia is key. If pathways that lead to collagen deposition and matrix remodeling (TGF- β , CAF activation, LOX crosslinking) are upregulated, the tumor physically shields itself. In immune-excluded TNBC biopsies, high collagen density (driven by CAFs) was noted, and T cells were stuck at the periphery ⁴⁹. An increase in fibrosis during therapy could indicate a wound-healing response that thwarts immune cell penetration. Therefore, *upregulation of pathways associated with ECM production* (e.g. TGF- β , PDGF, FAK signaling) signals resistance. This can be measured by a “CAF signature” – if that goes up, outcome is poor. Indeed, one analysis showed a **Turley CAF (cancer-associated fibroblast) gene signature** was higher in non-responders ⁵⁷. Therapies targeting these (e.g. enzymatically degrading collagen or inhibiting FAK) are being investigated to overcome this form of resistance.

Resistance Pathways for Downregulation: These are pathways where *loss or suppression* leads to resistance – often meaning a pathway that is needed for immune-mediated killing is turned off by the tumor:

- **MHC Class I Antigen Presentation (Down):** A classic acquired resistance mechanism to any T-cell-based therapy is loss of antigen presentation. Tumors can **downregulate MHC class I** by mutations in β 2-microglobulin (B2M) or antigen-processing machinery, or by epigenetic suppression of HLA genes. In breast cancer, B2M mutations are not common baseline, but there have been reports in advanced disease; one study noted an increase in B2M mutations in patients who had progressed on immunotherapy, suggesting immune editing ⁵⁸. If a tumor stops presenting antigens on MHC I, cytotoxic T-cells (even if activated by PD-L1 blockade) cannot recognize or kill the cancer cells – leading to treatment failure. Thus **downregulation of the antigen presentation pathway** is a major immune escape: e.g. loss of TAP1/TAP2 (peptide transporters), HLA heavy chain, or B2M. In a patient example outside breast cancer, acquired JAK1/2 mutations lead to lost IFN γ signaling and hence loss of MHC expression, causing resistance to PD-1 therapy – a concept likely extendable to TNBC ⁴⁹. For breast cancer specifically, evidence is still emerging, but the expectation is that tumors that initially respond may later recur with antigen presentation defects.
- **Interferon Signaling (Down):** Tumors might develop mutations in the IFN γ receptor or downstream JAK-STAT pathway. **JAK2-inactivating mutations** have been documented in melanoma resistance; in TNBC, one case report (Genomic profiling of a rapid progressor on neoadjuvant chemo + PD-1 blockade) found a JAK1 truncating mutation in a non-responding TNBC ⁵⁹. A **downregulation of the IFN γ pathway** in tumor cells means they no longer respond to T-cell-derived IFN γ – so they don't upregulate PD-L1 (making the drug moot) or MHC and death receptors. Essentially, IFN γ pathway loss shields tumor cells from the immune cytokine that is supposed to mark them for killing. Therefore, **loss of JAK/STAT signaling** is a resistance route. Similarly, loss of IFN α/β (type I IFN) sensing (STING or IRF3 pathway defects) would impede the priming of T-cells. In summary, if a tumor cell **downregulates interferon response genes**, it can hide from immune attack even if T-cells are present.

- **Tumor Neoantigen Expression (Down):** Tumors under immune pressure can undergo clonal evolution to lose the antigens that T-cells recognize. This is an immune-editing process rather than a canonical “pathway,” but it manifests as resistance. For example, if a specific neoantigen was driving the T-cell response and that subclone is eliminated, the remaining tumor might be ignored by T-cells. Technically, this is a loss of antigenic pathway. While hard to measure, it’s been observed in other cancers (e.g. loss of antigen in melanomas). In breast cancer, with lower mutation burden, this might be less common, but conceivable – especially in HR+ cancers where initial responses are rare.
- **Tumor Dependency on PD-L1 (Down):** Some tumors might simply not rely on PD-L1 for immune evasion. If a tumor lacks PD-L1 expression or downregulates it through some means (e.g. alterations in oncogenic pathways that control PD-L1 transcription like loss of PTEN can reduce PD-L1 in some contexts), then atezolizumab has no target. This is more of a primary resistance scenario: *PD-L1-negative tumors are resistant by default*. But interestingly, there are cases where a tumor that was PD-L1-positive becomes PD-L1-negative at progression (perhaps by outgrowth of a clone that doesn’t induce PD-L1). That is effectively a **loss of target** – though PD-L1-negative status usually indicates other immune evasion in play, like not having T-cells at all. Nonetheless, **downregulation of PD-L1 expression** after initial therapy could be an escape (since the drug only binds free PD-L1; if none is present, presumably the tumor has found another way to evade, or simply doesn’t face immune pressure because T-cells are exhausted by other means).
- **Co-stimulatory/Immune Cell Activation Pathways (Down):** Tumors may induce a state where co-stimulatory signals are lacking. For example, *downregulation of CD80/86 on dendritic cells* or loss of CD28 on T-cells will render T-cells anergic even if PD-L1 is blocked. While this is not well documented in breast cancer patients, the principle stands: if the immune synapse lacks activation signals, PD-L1 blockade alone can’t spark an immune attack. Chronic antigen exposure can cause T-cells to downregulate CD28 and rely on PD-1 pathways; blocking PD-1 gives them some function back, but eventually if they don’t get CD28 co-stimulation, they may fail. Thus, a microenvironment with, say, *downregulated IL-12 or CD27/CD28 signaling* could represent resistance – the T-cells are simply “paralyzed” in a different way.

In summary, **resistance mechanisms** often mirror the inverse of sensitivity ones. Tumors evade by *upregulating alternate brakes* (LAG-3, TIM-3, IDO, TGF- β , etc.) or by *downregulating critical immune process enablers* (MHC, IFN response). Breast tumors that progress on atezolizumab frequently show an immune profile shift: fewer TILs (immune desertification) and more stromal/suppressive signals compared to baseline ⁴⁵. This could be due to selection for resistant clones (e.g. MHC-loss variants) or adaptive resistance (e.g. recruitment of Tregs, MDSCs after therapy). Recognizing these pathways has important clinical implications: for instance, combining atezolizumab with a LAG-3 antibody (to counter LAG-3 upregulation), or with chemotherapy/radiation (to increase antigen release and IFN production) are strategies to circumvent these resistance routes. Each pathway we identified provides a target for rational combinations to **sustain or restore sensitivity** to immunotherapy.

Breast Cancer Subtype–Stratified Evidence

Breast cancer is heterogeneous, and atezolizumab's effects vary markedly across subtypes. Key subtype-specific evidence includes:

- **Triple-Negative Breast Cancer (TNBC):** This is the only subtype with proven clinical benefit from atezolizumab. TNBCs, especially basal-like, tend to have higher TILs and PD-L1 expression ³. The IMpassion130 trial that led to approval was in unresectable/metastatic TNBC; it showed that PD-L1⁺ TNBC derived a PFS (and a trend in OS) benefit ⁴ ⁶⁰. Within TNBC, PD-L1 positivity on immune cells ($\geq 1\%$ IC by SP142 assay) was required for efficacy ⁶. Further breakdown: *Basal-like immune-activated (BLIA)* TNBC responded best – characterized by high TILs, PD-L1, and low EMT/stroma ¹⁹. *Basal-like immune-suppressed (BLIS)* TNBC, which has high TGF- β and low immune infiltration, responded poorly. Post-hoc analyses suggested that patients with BRCA1/2-mutated or HRD⁺ TNBC (often overlapping with BLIA if they have TILs) might have numerically better outcomes, consistent with those tumors being more immunogenic ²⁸. Additionally, **spatial immune heterogeneity** in TNBC has been studied: tumors with immune cells dispersed inside tumor cell nests (immune-infiltrated) responded, whereas those with immune cells restricted to margins (immune-excluded) did not ⁴⁹. This links to the TGF- β /stromal discussion – an immune-excluded phenotype often corresponds to the BLIS molecular subtype.
- **HR+/HER2– (Luminal) Breast Cancer:** These tumors generally have low PD-L1 expression and low TIL levels. Clinical trials of checkpoint inhibitors in unselected HR+ metastatic breast cancer have been largely disappointing, with very low response rates (~0–5%). As such, atezolizumab is not indicated for HR+ disease. One reason is that luminal cancers tend to be “immune-cold”: they upregulate estrogen-driven proliferation pathways but not immune pathways. There is also evidence that endocrine therapy can modulate immune response; for instance, activation of the ER pathway may suppress some chemokine expression. Some trials are exploring immunotherapy in luminal cancers with high TMB or in combination with CDK4/6 inhibitors (since CDK4/6 inhibition can increase antigen presentation), but as of now, **no subtype-specific evidence supports atezolizumab use in HR+ breast cancer** beyond rare clinical trial anecdotes. In fact, the presence of HR expression might be somewhat immunosuppressive (through factors like TGF- β or IL-13 that correlate with luminal differentiation). Therefore, luminal breast cancer should be considered inherently resistant unless a high TMB or some extraordinary feature is present.
- **HER2-Positive Breast Cancer:** HER2+ tumors can have substantial immune infiltrates and PD-L1 expression, particularly in the HER2-enriched subtype and in cases with high TILs (which predict better response to HER2-targeted therapy). There is scientific rationale to combine checkpoint inhibitors with anti-HER2 therapy (since HER2 therapy can induce antibody-dependent cellular cytotoxicity and immune engagement). Small studies of atezolizumab or pembrolizumab added to HER2-directed regimens (like ado-trastuzumab emtansine or chemotherapy + trastuzumab) have shown some responses, but randomized data are limited. One phase 2 trial (KATE2: T-DM1 $\pm$ atezolizumab in metastatic HER2+ after progression) did not meet its primary PFS endpoint overall, **but in PD-L1+ subgroup a numerical PFS improvement was observed** ⁶¹. The sample size was small. Another study (NEOSTAR) in early HER2+ did not show a clear benefit of adding atezo to chemo+HER2 therapy in terms of pCR. Therefore, while HER2+ cancers can be immunogenic (especially if TIL-high), atezolizumab is *not standard* for this subtype; ongoing research aims to

identify if a subset (e.g. PD-L1+ and TIL-high HER2+) could benefit in the future. **Subtype nuance:** HER2-enriched (non-luminal) tumors have more immune activation than Luminal A/B HER2+ tumors. If any benefit exists, it would be in that group.

- **BRCA-Mutated/HRD+ Cancers:** As noted, a significant fraction of TNBC (and some luminal B) tumors have homologous recombination deficiency. gBRCA1/2-mutated TNBC often has a basal phenotype with high TILs. These patients might have heightened sensitivity to checkpoint blockade. For example, in a single-arm trial of pembrolizumab in metastatic TNBC, two out of the few responders were BRCA1-mutant (hypothesis-generating observation). Similarly, the presence of HRD was retrospectively associated with improved response in the MEDIOLA trial (durvalumab + PARP inhibitor in BRCA-mutant BC). While no definitive trial has stratified purely on HRD for immunotherapy in breast cancer, it is *suggestive* that **BRCA/HRD+ status contributes to an inflamed TME** (due to more neoantigens) and thus to atezolizumab efficacy ²⁸. One caution: not all HRD+ tumors have TILs (some HRD+ can still be immune cold if other factors like Wnt are in play). Nonetheless, HRD is an important subtype context to consider – it likely overlaps significantly with PD-L1 positivity in TNBC.
- **PD-L1-Negative TNBC:** It's worth delineating this "subtype" because even within TNBC, PD-L1 IHC-negative patients did not benefit in IMpassion130 (their PFS curves overlapped placebo) ⁴. These PD-L1-negative TNBCs might correspond to the BLIS subtype or simply have other means of immune evasion. They often have low TILs. This group highlights that not all TNBC is sensitive to immunotherapy – *only ~40% of TNBC that are PD-L1+ (by immune cell staining) show benefit*. PD-L1-negative TNBC represents an immune-cold subset that may require additional interventions (e.g., inducing inflammation via therapies like radiation or targeted therapy to render them PD-L1+ and inflamed).
- **Tumor Mutational Burden (TMB) High Subset:** Across breast cancer subtypes, a minority of tumors have very high TMB (often due to mismatch repair deficiency or APOBEC activity). Those with TMB ≥ 10 mutations/Mb may respond better to PD-1/L1 inhibitors; in fact, the FDA has an agnostic approval for pembrolizumab in TMB-high solid tumors. In breast cancer, this is rare (~5% of metastatic cases), but can occur in any subtype (though more in TNBC). It's not specifically studied for atezolizumab, but one could extrapolate that **TMB-high breast cancers** (regardless of ER/HER2) might have increased chance of response if they express PD-L1 or have TILs. Still, PD-L1 positivity and TIL presence remain more practical selectors in breast cancer, whereas TMB is not routinely used due to its low incidence and less clear cutoff in this disease.

In summary, **atezolizumab's benefit is largely confined to the immunologically "hot" subsets of breast cancer, predominantly PD-L1-positive TNBC** ⁴ ⁶⁰. Within that, basal-like immune-activated TNBC stands out as most responsive, while luminal/HER2-negative and immune-desert tumors are essentially unresponsive. HER2-positive disease has potential for immunotherapy benefit, but current evidence is insufficient for routine use. Stratifying patients by biomarkers like PD-L1, TILs, genomic instability, and immune gene signatures is essential in understanding and predicting who benefits from atezolizumab.

Contraindications and Safety

Atezolizumab, like other immune checkpoint inhibitors, has a distinct safety profile primarily involving immune-related adverse events. According to the FDA-approved label, **there are no absolute contraindications** to Tecentriq aside from known severe hypersensitivity to the drug or its components ⁶². However, the label and guidelines provide cautionary guidance in certain scenarios:

- **Immune-Mediated Conditions:** Patients with a history of severe autoimmune disease or prior immune-related adverse events may not be suitable candidates. While not a formal contraindication, clinicians exercise caution because unleashing the immune system could exacerbate underlying autoimmune disorders. The prescribing information notes that *atezolizumab should be avoided or used carefully in patients with active, severe immune-mediated conditions* such as **immune-mediated pneumonitis, colitis, hepatitis or endocrinopathies** ⁶³. In practice, if a patient has, for example, active Crohn's disease or lupus requiring immunosuppression, checkpoint inhibitors are typically not recommended due to risk of life-threatening flare.
- **Organ Transplant Recipients:** Although not explicitly listed in the label's contraindications, using PD-L1 inhibitors in solid organ transplant patients can provoke organ rejection (since the PD-1 pathway is important in graft tolerance). Case reports with other PD-1 inhibitors have shown transplant rejection. The atezolizumab label includes *solid organ transplant rejection* as a possible serious immune-mediated event ⁶⁴. Therefore, this scenario is generally avoided.
- **Pregnancy and Lactation:** Atezolizumab is classified as Pregnancy Category D (or the newer labeling indicates there is a risk to fetus). Checkpoint blockade can theoretically harm a fetus because the PD-1/PD-L1 pathway is involved in maintaining maternal-fetal tolerance ⁴⁸. Animal models have shown that PD-L1 or PD-1 knockout leads to higher rates of fetal loss ⁴⁸. The label explicitly advises verifying pregnancy status before starting atezolizumab and recommends effective contraception during treatment and for at least 5 months after the last dose ⁶⁵ ⁶⁶. Women should not breastfeed while on therapy, and for a few months after, due to potential harm to the infant (it is unknown if atezolizumab is secreted in breast milk, but IgG can cross into milk). Thus, **pregnancy is a de facto contraindication** and lactation is not recommended.
- **Infusion Reactions:** Severe infusion-related reactions (e.g., anaphylaxis) are rare but can occur. If a patient experiences a Grade 4 anaphylactic reaction to atezolizumab, re-challenge is contraindicated. The label mentions hypersensitivity reactions including anaphylaxis; facilities should have resuscitation equipment and epinephrine ready ⁶⁷ ⁶⁸. Patients with known hypersensitivity to Chinese Hamster Ovary cell products or any component of the formulation should not receive atezolizumab.
- **Severe Hepatic Impairment:** While not strictly contraindicated, patients with AST/ALT > 3×ULN (if due to liver metastases, >5×ULN) or baseline severe hepatic impairment (Child-Pugh C) were excluded from trials. Immune hepatitis is a concern. If baseline LFTs are extremely high, starting a hepatotoxic agent like an IO could be dangerous. So guidelines (e.g., ESMO, ASCO) suggest avoiding checkpoint inhibitors in uncontrolled hepatic autoimmune disease or acute hepatitis. If a patient has *active* HBV or HCV, treat those first (not a contraindication, but careful monitoring is needed as flares of viral hepatitis can occur when immune system is unleashed).

- **Severe Pneumonitis or ILD:** Patients with interstitial lung disease at baseline or a history of idiopathic pulmonary fibrosis could be at higher risk for immune pneumonitis. Again, not an absolute contraindication, but strong precaution. The label warns that immune-mediated pneumonitis can be fatal and should be managed with prompt steroids ⁶⁹. So if someone has pre-existing significant ILD requiring oxygen, using atezolizumab is very risky.
- **Others:** No contraindications for age or minor renal impairment exist. The elderly tolerate atezolizumab similarly to younger patients, per FDA review. Likewise, renal impairment is not a contraindication (atezo is not renally excreted significantly). There's no known drug-drug interaction that would contraindicate use, though caution is needed with systemic immunosuppressants (they could reduce efficacy; conversely, high-dose steroids might be needed to treat toxicity). The label doesn't list CYP interactions because it's a monoclonal antibody (not metabolized by liver enzymes in the same way).

In summary, **Tecentriq's formal contraindication list is essentially "None" aside from hypersensitivity** ⁷⁰, **but practical contraindications/precautions include:** active severe autoimmune disease, organ transplant, pregnancy, and uncontrolled intercurrent infections or conditions that could be worsened by immune activation. If any of the severe immune-related adverse events occur on treatment (e.g., Grade 4 pneumonitis, myocarditis, neurotoxicity), the drug must be permanently discontinued ⁷¹ ⁶⁹. These safety considerations are supported by regulatory labels (FDA/EMA) and clinical guidelines (NCCN advises against immunotherapy in patients with active autoimmune disease or organ transplant) ⁶³.

Trial and Guideline Context

Atezolizumab's development in breast cancer and its incorporation (and later removal) in guidelines is a story of initial promise followed by reevaluation:

- **Key Clinical Trials:** The seminal trial for breast cancer was **IMpassion130 (NCT02425891)** – a phase III trial in metastatic TNBC. It randomized patients to nab-paclitaxel with atezolizumab or placebo as first-line therapy. The trial met its primary PFS endpoint in the PD-L1-positive subgroup: median PFS 7.5 vs 5.0 months in PD-L1+ patients (HR 0.62, $P < 0.001$) ⁴ ⁶. An interim OS analysis showed a sizable improvement in PD-L1+ patients (25.0 vs 15.5 months, HR 0.62) though it was not formally statistically significant at that interim ⁷². This led to FDA granting **accelerated approval** in March 2019 for atezolizumab + nab-paclitaxel in PD-L1+ unresectable locally advanced or metastatic TNBC ⁵. The confirmatory trial was **IMpassion131 (NCT03125902)** – similar design but with paclitaxel (solvent-based) instead of nab-paclitaxel. Unfortunately, IMpassion131 *did not* show a PFS benefit (PFS was ~6 months in both arms, HR ~1.0) in PD-L1+ patients ⁷. No OS benefit was seen either, raising questions about the regimen differences or patient selection. In 2021, an FDA ODAC advisory panel, despite IMpassion131, voted to keep the TNBC indication given the positive data of IMpassion130 and lack of new safety issues ⁶¹. However, with new therapies emerging (e.g., pembrolizumab in TNBC with Keynote-355) and the confirmatory trial negative, Genentech voluntarily **withdrew the TNBC indication for atezolizumab in the US (Aug 2021)** ⁷³ ⁸. In Europe, EMA had also approved it; they too re-evaluated after the withdrawal but as of early 2022, atezo+nab-paclitaxel is generally no longer a preferred option in guidelines, replaced by pembrolizumab + chemotherapy (which got full approval based on positive OS in PD-L1 CPS ≥ 10).

- **Guidelines (NCCN/ESMO/ASCO):** NCCN (as of 2021) listed atezolizumab + nab-paclitaxel as an option for PD-L1 (IC $\geq 1\%$) metastatic TNBC, category 1 recommendation, prior to withdrawal. After withdrawal, NCCN notes this regimen is no longer FDA-approved, and currently recommends **pembrolizumab + chemo** for PD-L1 (CPS ≥ 10 by 22C3 assay) metastatic TNBC (which targets a slightly different PD-L1 metric) – effectively replacing atezo. ESMO's 5th Advanced Breast Cancer consensus (ABC5, 2020) stated: "PD-L1 status should be tested in first-line TNBC if immune checkpoint inhibitors are available" ⁷⁴ and recommended atezolizumab + nab-paclitaxel in PD-L1+ cases (level I, A evidence). Post-2021, ESMO guidelines note that atezolizumab is an option where approved, but highlight that pembrolizumab has level I evidence as well. ASCO guidelines (2022) similarly preferentially endorse pembrolizumab in PD-L1-positive TNBC, mentioning that atezolizumab is no longer an option in the US; however, they emphasize the principle that **immunotherapy combined with chemo is standard for PD-L1-positive TNBC** ⁷⁵. NICE in the UK approved atezolizumab+nab-paclitaxel in 2020 for PD-L1+ TNBC, but presumably will reassess given changes (NICE often re-evaluates if a drug loses regulatory approval elsewhere or if another cost-effective alternative emerges).
- **Other Trials:** There are ongoing studies of atezolizumab in *earlier-stage* TNBC. The **IMpassion031** trial (neoadjuvant atezo + chemo) showed an improvement in pathologic complete response (pCR) rates: 58% vs 41% in all-comers TNBC, regardless of PD-L1 status ⁷⁶. This led to an FDA approval in 2020 for pembrolizumab (not atezo) in the neoadjuvant TNBC setting; Genentech did not pursue an approval for atezo in neoadjuvant, perhaps due to focusing on metastatic disease at the time and then shifting priorities after withdrawal. Nonetheless, IMpassion031 indicated that atezolizumab can induce meaningful immune effects in early TNBC (higher pCR, especially in PD-L1+). Another adjuvant trial, **IMpassion030** (atezo after surgery and chemo in TNBC), was reported in 2023 and unfortunately was negative – no improvement in invasive disease-free survival was seen ⁷⁷. This dampened enthusiasm for adjuvant checkpoint blockade in TNBC outside of the neoadjuvant context.
- **Combinations and Niche Trials:** Atezolizumab is being tested in combination with various agents to overcome resistance pathways. For example, *TONIC trial* had an atezolizumab arm assessing induction strategies (like low-dose chemo or radiation to prime immune response). *BEGONIA* trial explores durvalumab (another PD-L1 inhibitor) with novel immunotherapy combos in TNBC – not directly atezo, but conceptually similar combos might be attempted with atezo in investigator-initiated studies. In HER2+ disease, *ATEZO-TRASTUZ* (single-arm atezo + trastuzumab in PD-L1+ HER2+ metastatic) showed a low response rate but durable disease control in a few, hinting at some activity in heavily pretreated HER2+ if PD-L1+.
- **Biomarker Insights from Trials:** From IMpassion130 biomarker sub-study: PD-L1 IC status was the key predictive biomarker ⁴. Neither tumor mutational burden nor BRCA status was prospectively tested, but retrospective analyses suggest no large difference by BRCA (though BRCA mutated tended to be PD-L1+ anyway). Gene expression analyses identified that an **"immune gene signature" was associated with better outcome** ²², whereas a **high TGF- β /stromal signature was associated with poorer outcome** ¹⁹. These reinforce that the presence of an immune-inflamed microenvironment is needed. Also, high baseline TILs, even in PD-L1-negative, might have slight benefit (but small numbers). Interestingly, the control arm patients who had high TILs did well with chemo alone too – immunotherapy mainly provided benefit on top of chemo in those whose tumors were actively suppressing immune response via PD-L1.

- **Current Status:** As of 2025, atezolizumab is no longer a recommended option for metastatic TNBC in the US (and many other regions) because pembrolizumab with chemo has shown OS benefit and atezo's confirmatory trial failed ⁷⁵. However, some patients who started atezo+nab-paclitaxel earlier and are doing well have continued it. In countries where nab-paclitaxel is not easily available or pembrolizumab not approved, atezolizumab might still be in use (for example, some Asian countries had ongoing access in 2022). The field has largely moved to using **pembrolizumab** in both early and metastatic TNBC where appropriate, given its broader indications (PD-L1 CPS ≥ 10 for metastatic; any PD-L1 for neoadjuvant). Yet, the scientific insights from atezolizumab trials remain very relevant – they illustrate the importance of immune phenotypes and have guided new combinations to address resistance (like adding TIGIT inhibitor *tiragolumab* to atezo in ongoing trials for TNBC, aiming to counter PD-L1 and another checkpoint simultaneously ⁷⁸).

In conclusion, **atezolizumab's journey in breast cancer has been one of initial groundbreaking approval in TNBC, followed by withdrawal in favor of alternative immunotherapy**. Guidelines now reflect this shift, emphasizing biomarker selection (PD-L1 or CPS, TIL counts) and combination approaches. The major trial evidence (IMpassion130/131, 031, 030) collectively suggests that immunotherapy can help a subset of breast cancer patients – particularly those with robust anti-tumor immune presence – but it is not universally effective. Ongoing studies and real-world data continue to refine the understanding of how to best integrate PD-L1 blockade into breast cancer care.

Other Mechanistic/Clinical Notes

- **PD-L1 Expression in Breast vs Other Cancers:** Breast cancers generally have lower PD-L1 positivity rates compared to say lung cancer. In TNBC, about 40–50% have PD-L1+ immune cells by the SP142 assay ⁷⁹. In HR+ or HER2+, the rates are lower. This partly explains why single-agent checkpoint inhibitors had modest success in breast cancer unless combined with chemotherapy (which can be immunogenic). Chemotherapy (like nab-paclitaxel) can increase antigen release and T-cell recruitment, which is why the combination was effective ⁸⁰ (nab-paclitaxel is thought to be more immunogenic than solvent-based paclitaxel, potentially due to less steroid premedication and more cell death patterns conducive to immunity).
- **Immune-Related Adverse Events (irAEs):** In breast cancer trials, the profile of irAEs on atezolizumab was similar to other cancers. Hypothyroidism (~10%), adrenal insufficiency (~<2%), pneumonitis (~2–3% any grade) ⁶⁹, and hepatitis (~1–2%) were observed. Notably, in IMpassion130, no new safety signals emerged; however, some concern arose about **neutropenia and infections** because combining with chemo could compound immunosuppression. Indeed, in IMpassion130 the rates of neutropenia, peripheral neuropathy, etc., were similar between arms (driven by chemo), but immune arms saw more hypothyroidism and skin rash. Immune toxicities are manageable with corticosteroids if recognized early. The risk/benefit in early breast cancer (curative setting) must weigh permanent endocrinopathies vs benefit – currently pembrolizumab in early TNBC is given with that caution.
- **Biomarker Assays:** It's worth noting the technical aspect: the trial defining PD-L1 positivity for atezo (Ventana SP142 assay on immune cells $\geq 1\%$) differs from the assay used in pembrolizumab trials (22C3 CPS ≥ 10). SP142 is known to stain fewer tumor cells and primarily immune cells; some tumors can be SP142-negative but CPS-positive and vice versa. There's been debate, but generally SP142 is

considered somewhat more restrictive. If applying those assays, one must use the specific cutoffs validated in trial for that drug. The fact that IMpassion131 failed raised questions if paclitaxel (which required steroid premedication) might dampen PD-L1 expression or immune activation compared to nab-paclitaxel – a hypothesis for why one taxane worked and not the other. Alternatively, it could have been a statistical fluke or subtle population difference.

- **Emerging Targets and Checkpoints:** The exploration of resistance led to interest in **TIGIT**, an inhibitory receptor on T cells and NK cells. Tiragolumab, an anti-TIGIT, is being combined with atezolizumab (since Roche developed both) in trials (e.g., CITYSCAPE trial in lung showed improved responses). In breast cancer, a trial called **SKYLINE** is evaluating atezolizumab + tiragolumab + chemo in front-line TNBC. This combination targets multiple checkpoints (PD-L1 and TIGIT) to reduce the chance of resistance via alternate pathways. Similarly, **LAG-3 inhibitors** (like relatlimab) could be paired in the future.
- **Microbiome and others:** Some research suggests the gut microbiome can influence immunotherapy efficacy. No breast-specific data yet, but given the general principle, broad-spectrum antibiotics near the start of atezo treatment might reduce efficacy (seen in melanoma/lung studies). Patients on chronic antibiotics or with dysbiosis could potentially respond less, although this is speculative in breast cancer.
- **Clinical Trial Context beyond Advanced Disease:** Atezolizumab is also being studied in *prevention of metastasis* or minimal residual disease contexts. For example, **A-Brave (NCT02926196)** is an ongoing trial of adjuvant atezolizumab in TNBC after standard therapy (somewhat like the failed IMpassion030, but possibly different patient subset). Results are pending but initial signals have not been very positive in pure adjuvant setting, emphasizing that neoadjuvant (where tumor is present to stimulate immunity) might be the key timing for immunotherapy in curative intent.
- **Head-to-Head vs Pembrolizumab:** There is no direct head-to-head trial of atezo vs pembro in TNBC. Indirectly, pembrolizumab (Keynote-355) showed both PFS and OS benefit in PD-L1 CPS ≥ 10 population, leading to full approval, something atezo didn't achieve due to confirmatory trial issues. Pembrolizumab's PD-L1 assay (CPS) captures more patients (roughly 50–60% of TNBC are CPS ≥ 10). It's believed that both drugs are similar in mechanism; differences in outcomes could be due to trial design or assay. Some patients who are CPS ≥ 10 might be SP142 IC $< 1\%$, meaning they qualify for pembro but not for atezo (when it was available). This discrepancy may have impacted results: Keynote-355's control arm did slightly worse than IMpassion130's, possibly because it included some patients who might not have benefitted from chemo alone but did with IO.
- **Special Populations:** Checkpoint inhibitors including atezolizumab have been used in patients with HIV on therapy, or chronic hepatitis, etc., in dedicated trials. A small study of atezolizumab in HIV+ cancer patients (all tumors, including some TNBC) found it to be feasible and safe if the virus is well-controlled. No unexpected infections occurred ⁸¹. This suggests that being HIV+ is not a contraindication if on ART with good CD4 counts, though data in breast cancer specifically is very limited.
- **Pathway Enrichment Data Integration:** The mechanistic understanding from gene set and pathway analyses (like those by Saleh et al. 2019 ²³ ²⁴ and the Roche poster ²⁷ ¹⁷) is being used to design “adaptive” trials. For example, if a patient's tumor biopsy on therapy shows rising Wnt

target gene expression or high TGF- β , some trials consider adding a Wnt inhibitor or TGF- β blocker dynamically.

In essence, atezolizumab opened the door for immunotherapy in breast cancer, and its story provides cautionary lessons (requiring confirmatory trials) and rich mechanistic insights that continue to influence breast cancer research. The drug is still very relevant in other cancers (bladder, lung, etc.), and cross-learning helps: e.g., from bladder cancer we know TGF- β gene signature strongly predicts failure of atezo, which was analogous to the TNBC findings ⁴⁹. As the field advances, combinations targeting multiple pathways identified above (perhaps “*atezolizumab + something*” becomes effective for some who didn’t respond to atezo alone) may expand the benefit of immunotherapy in breast cancer beyond the current subset.

Final Pathway Evidence Table

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
HALLMARK_INTERFERON_GAMMA_RESPONSE	Up	Sensitive	Upregulation of IFN- γ signaling indicates an ongoing immune attack; promotes MHC antigen presentation and T-cell recruitment, enhancing response to PD-L1 blockade. Tumors with high IFN- γ gene signatures respond better to atezolizumab ²² ³¹ .	²² ³¹

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
HALLMARK_ALLOGRAFT_REJECTION (T-cell activation)	Up	Sensitive	Represents cytotoxic immune infiltration (CD8 ⁺ T cells/NK cells active). High expression correlates with “inflamed” tumors; atezo unleashes these effector cells. Gene sets of T-cell cytotoxicity were enriched in responders ²² .	²² ¹⁰
KEGG PD-1/PD-L1 Checkpoint Pathway (hsa05235)	Up (PD-L1 present)	Sensitive	Presence of PD-L1 pathway activity is required for drug to act. PD-L1 upregulation (often by IFN γ) identifies tumors dependent on this checkpoint for immune escape; blocking it yields tumor regression ⁶ ⁵ .	⁶ ⁵
Wnt/β-Catenin Signaling	Up	Resistant	Wnt pathway activation in tumor or CAFs excludes T-cells and induces PD-L1 on stroma. High β -catenin signaling leads to “cold” tumors unresponsive to PD-L1 blockade. Wnt inhibition can restore sensitivity ³⁰ .	³⁰ ⁵⁰

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
TGF-β Signaling/EMT (e.g. REACTOME_TGFBETA_PATHWAY)	Up	Resistant	TGF- β drives dense stroma and immune exclusion. Tumors with high TGF- β and EMT (BLIS subtype) had poor response to atezolizumab ¹⁹ . Downregulating TGF- β (e.g. in “immune-infiltrated” tumors) is associated with better response.	¹⁹ ⁴⁹
Antigen Presentation (GO:0019882)	Down	Resistant	Loss of MHC class I presentation (via B2M or JAK mutations) lets tumors hide from T-cells. If this pathway is downregulated, T-cells can’t recognize tumor even if activated. B2M mutations and lack of HLA observed in resistant cases ⁵⁸ .	⁵⁸ ⁴⁹

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
LAG-3 Checkpoint Pathway	Up	Resistant	Compensatory upregulation of LAG3 on T-cells provides an alternative brake when PD-L1 is blocked. TNBC studies show PD-1/L1 blockade increases LAG-3/TIM-3 on Tregs/CD4 T cells ⁵³ . LAG-3 high expression is linked to exhausted T-cells and non-response.	⁵⁴ ⁵³
PI3K/AKT/mTOR Pathway (e.g. HALLMARK_PI3K_AKT_MTOR_SIGNALING)	Up	Resistant	Oncogenic PI3K signaling fosters an immune-suppressive milieu (via VEGF, cytokines) and reduces TILs. PTEN loss (↑ PI3K) in TNBC correlates with poor ICI response ²⁹ . Inhibition of AKT can re-sensitize tumors to atezo.	²⁹ ⁸²

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
Hypoxia/Angiogenesis (HALLMARK_HYPOXIA)	Up	Resistant	Hypoxic tumors (high HIF-1 α , VEGF) inhibit effector immune function and T-cell infiltration. High hypoxia gene signature correlates with immune exclusion and atezo-resistance. Atezo-treated responders showed reduced hypoxia pathways ²⁵ .	²⁵ ¹⁷
DNA Repair Deficiency (HRD, e.g. KEGG Homologous Recombination hsa03440)	Down (deficient)	Sensitive	HR-deficient tumors have high mutation load and neoantigens, leading to TIL-rich, PD-L1+ environment ²⁸ . BRCA1/2-mutated TNBC respond well as their HR pathway is down, yielding more immunogenicity. (Conversely, on-treatment upregulation of DNA repair is a tumor escape response).	²⁸ ²⁵

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
NF-κB-Mediated Inflammation (TNFα via NF-κB)	Up	Sensitive	Pro-inflammatory signaling in the tumor microenvironment (via NF-κB) leads to chemokine release and dendritic cell activation. Tumors with active NF-κB pathways (producing CXCL9/10, etc.) recruit T-cells and respond better ¹⁷ . However, intrinsic NF-κB survival signals can also promote resistance if unchecked. Overall, an <i>immune</i> NF-κB profile (high cytokines) is favorable.	¹⁷ ²⁴
IDO-Kynurenine Pathway	Up	Resistant	IDO catabolizes tryptophan to kynurenine, suppressing T-cell proliferation. Tumors that induce IDO (often via IFNγ) dampen the immune response even with PD-L1 blocked. High IDO1 in TNBC correlates with poor response; IDO inhibition can improve T-cell activity ⁵² .	⁵²

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
Co-stimulatory signaling (CD28/B7 & IL-2)	Down	Resistant	If co-stimulation is lacking (e.g. low CD80/86 on APCs, low IL-2 signaling), T-cells cannot fully activate. Even with PD-L1 blocked, T-cells may remain anergic. A tumor microenvironment that downregulates these pathways (perhaps via lack of dendritic cells or presence of corticosteroids) will resist immunotherapy. Clinical data in breast are sparse, but this principle is inferred from immunology.	54 46

Pathway (ID/Name)	Regulation	Effect of Drug	Biological Rationale	References
Treg/Myeloid Immunosuppressive Pathways	Up	Resistant	An increase in Treg and MDSC-associated pathways (e.g. high CCL2, IL-6, ARG1) leads to accumulation of suppressive cells that inhibit effector T-cells. Atezolizumab efficacy is reduced in tumors with high FoxP3 ⁺ Tregs or CD11b ⁺ myeloid infiltration. For instance, elevation of IL-6 and IL-8 in plasma was associated with poorer outcomes (seen in other solid tumors). Reducing these pathways (via downregulation or targeted agents) is thought to improve response.	49 45

Legend: *Up* = pathway is increased in activity/expression; *Down* = pathway is decreased/suppressed. *Effect of drug (Sensitive/Resistant)* denotes whether the pathway's modulation leads to tumor being more sensitive or resistant to atezolizumab. Each pathway entry is supported by breast cancer-specific evidence or, where breast-specific evidence is limited, strong mechanistic rationale from immuno-oncology contexts. The references include clinical trial data (e.g. gene signatures from IMpassion130) and reputable pathway databases (KEGG, GO) or studies, illustrating how these pathways interplay with PD-L1 blockade in breast cancer.

2 21 KEGG DRUG: Atezolizumab

<https://www.kegg.jp/entry/D10773>

3 Atezolizumab in metastatic triple-negative breast cancer - ESMO Open

[https://www.esmoopen.com/article/S2059-7029\(20\)32770-8/fulltext](https://www.esmoopen.com/article/S2059-7029(20)32770-8/fulltext)

4 6 10 60 72 79 80 Atezolizumab and Nab-Paclitaxel in Advanced Triple-Negative Breast Cancer - PubMed

<https://pubmed.ncbi.nlm.nih.gov/30345906/>

5 7 8 61 Roche Withdraws Atezolizumab PD-L1-Positive Metastatic TNBC Indication in the United States | OnLive

<https://www.onclive.com/view/roche-withdraws-atezolizumab-pd-l1-positive-metastatic-tnbc-indication-in-the-united-states>

9 75 Still the Standard of Care for Metastatic Triple-Negative Breast Cancer

<https://dailynews.ascopubs.org/do/chemoimmunotherapy-still-standard-care-metastatic-triple-negative-breast-cancer>

11 Compound: ATEZOLIZUMAB (ChEMBL3707227) - ChEMBL

<https://www.ebi.ac.uk/chembl/explore/compound/ChEMBL3707227>

15 16 20 atezolizumab | Ligand page | IUPHAR/BPS Guide to PHARMACOLOGY

<https://www.guidetopharmacology.org/GRAC/LigandDisplayForward?ligandId=7990>

17 18 22 27 31 32 33 34 35 40 42 43 44 45 57 Mechanisms of action and acquired resistance to atezolizumab plus nab-paclitaxel in metastatic triple negative breast cancer (mTNBC)

<https://www.cuba.dialogoroche.com/content/dam/roche-dialogo/global-assets/downloadable-assets/congresos-y-eventos/oncology/roche-asco-2022/07-oncology-annual-meeting-2022-poster-molinero-mechanisms-of-action-and-acquired-resistance-to-atezolizumab.pdf>

19 [PDF] The Tumor Microenvironment and Atezolizumab + nab-Paclitaxel ...

<https://www.cuba.dialogoroche.com/content/dam/roche-dialogo/global-assets/downloadable-assets/oncology/roche-asco-2021/presentaciones/oncology-annual-meeting-2021-presentation-emens-the-tumor-microenvironment-TME-and-atezolizumab-nab-paclitaxel-A-nP-activity-in-metastatic-triple-negative-breast-cancer.pdf>

23 24 25 26 37 38 39 41 47 PD-L1 Blockade by Atezolizumab Downregulates Signaling Pathways Associated with Tumor Growth, Metastasis, and Hypoxia in Human Triple Negative Breast Cancer

<https://www.mdpi.com/2072-6694/11/8/1050>

28 Homologous Recombination Repair Deficiency and the Immune ...

<https://www.sciencedirect.com/science/article/pii/S1936523319304954>

29 82 First-Line Ipatasertib, Atezolizumab, and Taxane Triplet for ...

<https://aacrjournals.org/clincancerres/article/30/4/767/734225/First-Line-Ipatasertib-Atezolizumab-and-Taxane>

30 51 55 Frontiers | Wnt Inhibition Sensitizes PD-L1 Blockade Therapy by Overcoming Bone Marrow-Derived Myofibroblasts-Mediated Immune Resistance in Tumors

<https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2021.619209/full>

36 PD-L1 Blockade by Atezolizumab Downregulates Signaling Pathways Associated with Tumor Growth, Metastasis, and Hypoxia in Human Triple Negative Breast Cancer - PubMed

<https://pubmed.ncbi.nlm.nih.gov/31349612/>

46 The role of tumor microenvironment and immune cell crosstalk in ...

<https://www.sciencedirect.com/science/article/pii/S030438352500432X>

48 62 64 65 66 67 68 69 70 71 81 DailyMed - TECENTRIQ- atezolizumab injection, solution
<https://dailymed.nlm.nih.gov/dailymed/lookup.cfm?setid=6fa682c9-a312-4932-9831-f286908660ee>

49 Recent progress in immune evasion mechanisms of triple-negative ...
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12624998/>

50 WNT signaling modulates PD-L1 expression in the stem cell ...
<https://www.nature.com/articles/s41388-019-0700-2>

52 IDO Inhibition Facilitates Antitumor Immunity of Vy9Vδ2 T Cells in ...
<https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2021.679517/full>

53 54 Mechanisms and Strategies to Overcome PD-1/PD-L1 Blockade ...
<https://www.mdpi.com/2072-6694/15/1/104>

56 Combinatorial immunotherapies overcome MYC-driven immune ...
<https://www.nature.com/articles/s41467-022-31238-y>

58 Immune-related biomarkers in triple-negative breast cancer
<https://link.springer.com/article/10.1007/s12282-021-01247-8>

59 Genomic and Immune Profiling of a Patient With Triple-Negative ...
<https://ascopubs.org/doi/10.1200/PO.18.00335>

73 Genentech Withdraws Breast Cancer Indication From Tecentriq
<https://www.breastcancer.org/research-news/genentech-withdraws-breast-cancer-indication-from-tecentriq>

74 5th ESO-ESMO international consensus guidelines for advanced ...
[https://www.annalsofoncology.org/article/S0923-7534\(20\)42460-3/fulltext](https://www.annalsofoncology.org/article/S0923-7534(20)42460-3/fulltext)

76 Peri-operative atezolizumab in early-stage triple-negative breast ...
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12283416/>

77 Adjuvant Atezolizumab Plus Chemo Does Not Add Survival Benefit ...
<https://www.cancernetwork.com/view/adjuvant-atezolizumab-plus-chemo-does-not-add-survival-benefit-in-tnbc>

78 NCT06175390 | Tiragolumab, Atezolizumab and Chemotherapy in ...
<https://clinicaltrials.gov/study/NCT06175390>